Chapter 10. Linear Regression

Least squares, permutation and bootstrap inference, robust regression and M-estimation

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10.1 Simple linear regression

Model $y_i = \alpha + \beta x_i + \epsilon_i$ with uncorrelated errors of mean 0 and variance σ^2; inference is conditional on the x_i. Least squares gives $\hat\beta = S_{xy}/S_{xx}$, $\hat\alpha = \bar y - \hat\beta\bar x$ and $s^2 = \sum e_i^2/(n-2)$.

Proposition 10.1

$E\hat\beta = \beta$ and $\operatorname{Var}\hat\beta = \sigma^2/S_{xx}$. The residuals satisfy $\operatorname{Var}(e_i) = \sigma^2(1-h_i)$ with leverage $h_i = \frac{1}{n} + \frac{(x_i-\bar x)^2}{S_{xx}}$, so the adjusted residuals $r_i = e_i/\sqrt{1-h_i}$ have variance σ^2.

Proof. $\hat\beta = \sum w_i y_i$ with $w_i = (x_i - \bar x)/S_{xx}$, $\sum w_i = 0$, $\sum w_i x_i = 1$, $\sum w_i^2 = 1/S_{xx}$. With the hat matrix H, $\mathbf{e} = (I-H)\boldsymbol\epsilon$ and $\operatorname{Var}\mathbf{e} = \sigma^2(I-H)$, whose diagonal entries are $\sigma^2(1-h_i)$. ■

Under normal errors, $(\hat\beta - \beta)/\widehat{\mathrm{se}}(\hat\beta) \sim t_{n-2}$ with $\widehat{\mathrm{se}}(\hat\beta) = s/\sqrt{S_{xx}}$. Without normality we use resampling.

Permutation test of $H_0 : \beta = 0$. Under H_0, $y_i = \alpha + \epsilon_i$ are exchangeable, so shuffle the y's against the fixed x's. The statistics $\hat\beta$, $\sum x_i y_i$ and $t = \hat\beta/\widehat{\mathrm{se}}(\hat\beta)$ give the same test: with S_{xx} and S_{yy} fixed, $t = \hat\beta\sqrt{(n-2)S_{xx}}/\sqrt{S_{yy} - \hat\beta^2 S_{xx}}$ is increasing in $\hat\beta$. This is the same test as the permutation test of correlation (Section 5.1).

Bootstrap intervals. Use the case bootstrap (random x, robust to model failure) or the residual bootstrap $y_i^* = \hat\alpha + \hat\beta x_i + r_i^*$ (fixed design, model assumed). Because $\widehat{\mathrm{se}}(\hat\beta)$ has a formula, the bootstrap-t interval

$$\beta \in \big(\hat\beta - \widehat{\mathrm{se}}(\hat\beta)Q_{t^*}(1-\tfrac{\alpha}{2}),\ \hat\beta - \widehat{\mathrm{se}}(\hat\beta)Q_{t^*}(\tfrac{\alpha}{2})\big), \qquad t^* = \frac{\hat\beta^* - \hat\beta}{\widehat{\mathrm{se}}(\hat\beta^*)},$$

is cheap and second-order accurate. For tests of $\beta = 0$:

- null resampling: resample x and y *independently* (case form), or $y^* = \bar y + r_0^*$ with $r_{0,i} = (y_i - \bar y)/\sqrt{1-1/n}$ (residual form);
- pivot resampling: use t^* above and compare with $t_{\mathrm{obs}} = \hat\beta/\widehat{\mathrm{se}}(\hat\beta)$, which inverts the bootstrap-t interval.

10.2 Multiple regression

$\mathbf{y} = X\boldsymbol\beta + \boldsymbol\epsilon$ with X of size $n \times (p+1)$: $\hat{\boldsymbol\beta} = (X'X)^{-1}X'\mathbf{y}$, $E\hat{\boldsymbol\beta} = \boldsymbol\beta$, $\operatorname{Var}\hat{\boldsymbol\beta} = \sigma^2(X'X)^{-1}$, $s^2 = \mathbf{e}'\mathbf{e}/(n-p-1)$ (unbiased since $E\,\boldsymbol\epsilon'(I-H)\boldsymbol\epsilon = \sigma^2\operatorname{tr}(I-H)$). Each β_j is a *partial* effect: the effect of x_j adjusted for the other regressors.

Comparing nested models. Split $X = [X_0, X_1]$ ($k+1$ and $q = p-k$ columns) and test $H_0 : \boldsymbol\beta_1 = 0$. With hat matrices H and H_0,

$$F = \frac{\mathbf{y}'(H - H_0)\mathbf{y}/q}{\mathbf{y}'(I-H)\mathbf{y}/(n-p-1)} = \frac{\hat{\boldsymbol\beta}_1'\{X_1'(I-H_0)X_1\}\hat{\boldsymbol\beta}_1}{q\,s^2}. \tag{10.1}$$

Proposition 10.2

Under H_0 with normal errors, $F \sim F_{q,n-p-1}$. More generally $F_{\boldsymbol{\beta}_1}$, defined by replacing $\hat{\boldsymbol{\beta}}_1$ with $\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_1$ in (10.1), has this distribution for every $\boldsymbol{\beta}_1$, so it is a pivot.

Proof. $H - H_0$ is a projection of rank q orthogonal to $I - H$, and both annihilate X_0. Under H_0, the numerator and denominator quadratic forms reduce to $\boldsymbol{\epsilon}'(H-H_0)\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon}'(I-H)\boldsymbol{\epsilon}$, independent $\sigma^2\chi^2_q$ and $\sigma^2\chi^2_{n-p-1}$ (Cochran). The second form of (10.1) is Frisch–Waugh–Lovell: $\hat{\boldsymbol{\beta}}_1$ is the regression of $(I - H_0)\mathbf{y}$ on $\tilde{X}_1 = (I - H_0)X_1$, and $H - H_0$ projects onto the column space of $\tilde{X}_1$. For general $\boldsymbol{\beta}_1$, apply the same argument to $\mathbf{y} - X_1\boldsymbol{\beta}_1$. ■

For $q = 1$, $F = t_j^2$.

Permutation tests for $H_0 : \boldsymbol{\beta}_1 = 0$. Under H_0 the errors are exchangeable but the y_i are not (their means depend on X_0), so no permutation test is exact. Three approximate schemes:

1. *Shuffle y:* effectively tests $\beta_1 = \cdots = \beta_p = 0$; it can be anticonservative and is not recommended.
2. *Freedman–Lane:* fit the reduced model, shuffle its residuals e_0, set $y^* = \hat{y}_0 + e_0^*$, and recompute F. This is generally the most accurate.
3. *ter Braak:* shuffle the full-model residuals, set $y^* = \hat{y} + e^*$, and recompute the pivot $F_{\hat{\boldsymbol{\beta}}_1}$.

All three are asymptotically equivalent. Replacing "shuffle" with "resample with replacement" in schemes 2 and 3 gives the residual bootstrap tests under the null and with the pivot, respectively. Bootstrap-t intervals for each β_j use $t_j^* = (\hat{\beta}_j^* - \hat{\beta}_j)/\widehat{\mathrm{se}}(\hat{\beta}_j^*)$ with $\widehat{\mathrm{se}}^2(\hat{\beta}_j) = s^2(X'X)^{-1}_{jj}$.

10.3 Robust regression

Least squares is the maximum likelihood estimator under normal errors. It loses efficiency under heavy-tailed errors and can be ruined by a single outlier.

Definition (Breakdown point)

The smallest fraction of observations that, sent to infinity, can send the estimator to infinity. Roughly: the mean has $1/n \to 0$, the α-trimmed mean α, and the median $1/2$.

10.3.1 M-estimators

For location, the least squares estimator $\bar{x} = \arg\min_\mu \sum (x_i - \mu)^2$ is generalized to

$$\hat{\mu}^\Psi = \arg\min_\mu \sum_i \Psi\Big(\frac{x_i - \mu}{\hat{\tau}}\Big) \iff \sum_i \psi\Big(\frac{x_i - \mu}{\hat{\tau}}\Big) = 0, \qquad \psi = \Psi',$$

with a robust scale $\hat{\tau} = 1.4826$ MAD (consistent for σ under normality, since $\Phi^{-1}(3/4) = 0.6745$).

- $\psi(x) = x$: the mean. $\psi(x) = \mathrm{sgn}(x)$: the median.
- **Huber:** $\psi(x) = \max(-k, \min(k, x))$, quadratic Ψ in the center and linear in the tails. With $k \approx 1.35$ it has about 95% efficiency at the normal.
- **Tukey biweight:** $\psi(x) = x(1 - (x/c)^2)^2 I(|x| \le c)$, redescending, so gross outliers get zero weight.

A bounded ψ with a 50% breakdown scale gives breakdown point 50%. In regression, $\hat{\boldsymbol{\beta}}^{\Psi} = \arg\min \sum \Psi((y_i - \mathbf{x}_i'\boldsymbol{\beta})/\hat{\tau})$. The estimating equations $\sum \psi(\cdot)\,\mathbf{x}_i = 0$ show that Huber's estimator is robust to outlying y but not to outlying x (high leverage), because the factor x_{ij} is unbounded. **Yohai's MM-estimator** uses a redescending ψ (biweight, $c = 4.685$ for 95% efficiency) started from a high-breakdown S-estimate of scale, achieving a 50% breakdown point together with high efficiency. **Least trimmed squares** minimizes the sum of the h smallest squared residuals ($h \approx n/2$ for maximal breakdown); it is very robust but inefficient.

10.3.2 The Theil–Sen estimator

For $y_i = \alpha + \beta x_i + \epsilon_i$, form the pairwise slopes $\beta_{ij} = (y_j - y_i)/(x_j - x_i)$ for $x_i \neq x_j$ and set

$$\hat{\beta}^{TS} = \text{median}_{i<j}\, \beta_{ij}, \qquad \hat{\alpha}^{TS} = \text{median}_i (y_i - \hat{\beta}^{TS} x_i).$$

Proposition 10.3 (Link to Kendall's tau)

Let $\tau_n(\beta_0)$ be Kendall's statistic between x_i and $y_i - \beta_0 x_i$. Then $\binom{n}{2}\tau_n(\beta_0) = \#\{\beta_{ij} > \beta_0\} - \#\{\beta_{ij} < \beta_0\}$. Hence the test of $H_0 : \beta = \beta_0$ by Kendall's tau, and the interval $(\beta_{(l)}, \beta_{(u)})$ with $l = \binom{n}{2}\frac{Q_K(\alpha/2)+1}{2}$, $u = \binom{n}{2}\frac{Q_K(1-\alpha/2)+1}{2}$ (Q_K the null quantile of τ_n), are exact. $\hat{\beta}^{TS}$ is the corresponding point estimate.

Proof. $\text{sgn}[(x_i - x_j)\{(y_i - \beta_0 x_i) - (y_j - \beta_0 x_j)\}] = \text{sgn}[(x_i - x_j)^2(\beta_{ij} - \beta_0)] = \text{sgn}(\beta_{ij} - \beta_0)$. At the true β, $y_i - \beta x_i = \alpha + \epsilon_i$ is independent of x_i, so $\tau_n(\beta)$ has the null law K_n. The acceptance region $Q_K(\frac{\alpha}{2}) \le \tau_n(\beta_0) \le Q_K(1 - \frac{\alpha}{2})$ is a range of $\#\{\beta_{ij} > \beta_0\}$, which is a step function of β_0 dropping by one at each ordered slope. So the region is an interval of ordered slopes with the stated indices. ■

$\hat{\beta}^{TS}$ has breakdown point $1 - 1/\sqrt{2} \approx 29\%$ and about 96% ARE relative to least squares under normal errors.

Example 10.1

Points $(1,1), (2,2), (3,3.5), (4,4), (5,20)$, where the last point is an outlier. The ten ordered pairwise slopes are $0.5, 1, 1, 1, 1.25, 1.5, 4.75, 6, 8.25, 16$, so $\hat{\beta}^{TS} = 1.375$ and $\hat{\alpha}^{TS} = -0.625$. Least squares gives slope 4.0, pulled up by a single point.

In multiple regression Theil–Sen has no unique definition; one version iterates single-variable Theil–Sen fits by backfitting. Inference for robust estimators is usually done by bootstrap (bootstrap-t with the approximate standard errors from the final iteratively reweighted least squares fit) or by permutation tests of the studentized coefficients.